



# OPEN Safety risk analysis of high dosage of phosphogypsum in limestone soil and yellow soil: a case study of potted amaranth

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The safe management and sustainable utilization of phosphogypsum (PG) tailings in Guizhou, China, represent a pressing issue. High dosage incorporation of PG into karst soil has emerged as a promising approach. This method not only addresses the long-term storage issue of substantial PG tailings but also has the potential to enhance soil fertility. However, its safety risks to soil and food chain remain unclear. Through pot experiments, we incorporated 10–20% PG into limestone soil and yellow soil from the Karst region. The findings indicated significant increases in soil organic carbon (from 4.12% to 7.87%), and total salts (from 2.80 g/kg to 36.65 g/kg). Additionally, the concentration of potentially toxic elements (PTEs) in the soil did not exceed the established safety standards. The dry weight of *Amaranthus tricolor* showed a notable increase (from 0.01 g/plant to 0.78 g/plant), although the concentrations of Cu, Zn, and Cr exceeded permissible limits. Furthermore, the pH of the PG-added yellow soil increased significantly (from 5.40 to 6.03), whereas the pH of the limestone soil decreased notably (from 7.18 to 6.93). The microbial diversity analysis revealed PG had a more profound impact on the structural changes in bacterial and fungal communities within the yellow soil compared to the limestone soil, significantly influenced bacterial community composition, reducing saprophytic fungi and plant pathogens within the fungal community. The relative abundance of Bacteroidota and Firmicutes increased significantly in the yellow soil samples, while that of Proteobacteria and Acidobacteria decreased notably. In conclusion, high dosage PG utilization has the potential to enhance soil nutrients and optimize microbial community structures, thereby promoting plant growth. However, it may also lead to PTEs' accumulation, posing potential risks to food chain and ecological security. The study provides basic data for evaluating the environmental and ecological impacts of high dosage PG utilization in karst soils.

**Keywords** Phosphogypsum, Soilization, Solid waste utilization, Pot experiment, Environmental effects

The phosphate fertilizer industry plays a crucial role in ensuring global food security<sup>1</sup>. However, large-scale production of phosphate fertilizers generates substantial amounts of phosphogypsum (PG) as a by-product, which poses notable environmental risks to surrounding ecosystems<sup>2,3</sup>. PG is a solid waste produced in the wet-process phosphoric acid production, primarily through the reaction of sulfuric acid with phosphate rock. Its mineral and chemical composition are rather complex. Besides calcium sulfate (mainly calcium sulfate dihydrate, typically  $\geq 90\%$  by content and structurally identical to natural gypsum), it also contains undecomposed apatite, quartz, residual phosphoric acid, fluorides, acid-insoluble substances, organic matter, etc. Due to the high cost and technical challenges associated with PG treatment, land stacking remains the primary disposal method globally, supplemented by partial ocean dumping and industrial utilization<sup>4</sup>. Currently, global PG accumulation has exceeded 6 billion tons, with approximately 100 to 280 million tons added annually<sup>5–7</sup>. However, only about 15% of PG is recycled<sup>8,9</sup>. In China, cumulative PG production has surpassed 500 million tons, with an annual growth rate exceeding 50 million tons, primarily concentrated in regions such as Guizhou, Yunnan, Sichuan, Hubei, and Hunan. This rapid accumulation of PG leads to significant land occupation, plus its comprehensive

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utilization rate is below 40%, posing substantial challenges to the sustainable development of China's phosphate fertilizer industry<sup>10</sup>. Pollutants from PG infiltrate the terrestrial ecosystems through biological and chemical weathering processes<sup>11</sup>. These pollutants enter the ecosystem via wind dispersal, rainwater erosion, and groundwater infiltration, leading to widespread contamination of water, soil, and air within several kilometers of PG dumping sites with potentially toxic elements (PTEs) and other pollutants<sup>12</sup>. To address these challenges, exploring alternative applications for PG and enhancing its utilization efficiency is imperative. Additionally, strategies must be developed to manage the vast quantity of PG and mitigate its associated environmental contamination risks.

PG is an acidic mixture primarily consisting of calcium sulfate (pH 2–5), along with PTEs (Pb, Cd, Cu, Zn, Cr, As, etc.), organic matter (e.g., flocculants and flotation agents), radioactive elements (<sup>238</sup>U, <sup>210</sup>Pb, <sup>226</sup>Ra, <sup>232</sup>Th, etc.), rare earth elements (La, Ce, Gd, etc.), and nutrients (S, Ca, P, Fe, K, etc.)<sup>13</sup>. It is a gray, light-gray, or light-yellow fine powder and is utilized in various sectors such as construction materials, chemical industry, medical facilities, rare earth recovery, carbon fixation, desert fixation, and sewage treatment<sup>14–18</sup>. Nevertheless, the widespread application of PG has encountered three principal constraints: technological bottlenecks in implementation, economic viability concerns, and ecological sustainability issues, with rapid deployment at scale over short timeframe proving especially problematic<sup>19</sup>. Despite these limitations, recent studies highlight the potential of PG in agricultural and forestry applications<sup>20</sup>. Its acidic characteristics and fertilizer properties can be utilized as soil conditioners to improve saline-alkali land<sup>21</sup>, enhancing soil permeability, organic matter content, bulk density, salinity, alkalinity, and biological stress levels after remediation<sup>22</sup>. PG also has the potential to mitigate Al<sup>3+</sup> toxicity through coprecipitation and adsorption in acidic soils without significantly affecting soil pH<sup>23</sup>. Additionally, PG can potentially immobilize PTE pollution through ion exchange, adsorption, and the formation of secondary minerals, facilitating the remediation of contaminated soil<sup>24</sup>. Given the extensive area of normal cultivated land and optimal PG quantities as fertilizer or soil amendment, it can provide essential nutrients (Ca, K, S, and P) critical for crop production<sup>20,25–27</sup> while maintaining acceptable PTE and radiation levels in soil<sup>20</sup>. Although current PG soilization can utilize a portion of the accumulated waste, the amount of PG added is often relatively low. For instance, in pot experiments, the addition rate typically does not exceed 1%<sup>28,29</sup>, while in field experiments, it generally remains below 5 t/ha<sup>30,31</sup>. Such low application rates are inadequate for utilizing the large and continuously growing stockpiles of PG. High dosage PG utilization presents a promising approach to these challenges. Guizhou Province, as a mountainous region with the largest contiguous karst landscapes in China, features thin soil layers, severe soil erosion, and a fragile ecological environment. Its primary soil types—limestone soil and yellow soil—exhibiting notable differences in origin, composition, and fertility characteristics<sup>32,33</sup>, the region serves as an ideal testing ground for evaluating the effectiveness of high-dose PG utilization.

Therefore, this study aims to investigate the effects of high dosage PG incorporation (10% and 20%) into karst soil on plant growth, soil physicochemical properties, and soil microbial community structure by pot experiments. *Amaranthus tricolor*, a widely cultivated economic plant with a short growth cycle, strong environmental adaptability, and high mineral weathering capability, was selected as the test species. Its ability to accumulate PTEs enables an assessment of the potential risks of PTE contamination associated with high dosage PG application<sup>34</sup>. These characteristics make it a suitable model plant for pot experiments. The findings will provide basic information for evaluating the ecological and environmental impacts of large-scale and high dosage PG utilization in karst mountainous regions.

Materials and methods  
Raw materials and pot experiment design

The predominant soil types in Guizhou Province—yellow soil and limestone soil—were collected from Puding County, a region characterized by extensive karst development in China. After collection, the soils were air-dried, cleared of plant roots and gravel, ground to pass through a 60-mesh sieve and stored for subsequent use. The PG utilized in the experiment was obtained from the Hetaoqing PG disposal site in Xifeng County, Guizhou Province. It was similarly air-dried and preserved for subsequent use. The composition of PG was determined using a X-ray fluorescence spectrometer (Shimadzu EDX-7000), and the results are presented in Table 1. The organic matter content in PG was measured by the loss on ignition method (according to Chinese National Standard HJ 761–2015), revealing an organic matter content of approximately 0.3%. The particle size distribution was analyzed with a laser particle size analyzer (Betttersize 2600), indicating an average particle size of 94.39 μm. After dissolving PG in ultrapure water at a solid-to-liquid ratio of 1:10, the pH of the supernatant was measured to be 4.14. The pot design featured a lower diameter of 124 mm, an upper outer diameter of 152 mm, and a height of 119 mm. SiO<sub>2</sub> was purchased from Sinopharm Chemical Reagent Co., Ltd., China. The specific experimental treatments are summarized in Table 2.

On March 28, 2024, seeds of *A. tricolor* were sown, with approximately six seeds placed in each pot. The potted plants were placed in an open and ventilated rainproof greenhouse, with plant growth conditions maintained consistent with the surrounding environment. Daily irrigation was carried out using deionized water to maintain soil moisture at approximately 65% of the field water capacity. After emergence, five plants were retained per pot

|                   | SO <sub>3</sub> | CaO   | SiO <sub>2</sub> | Al <sub>2</sub> O <sub>3</sub> | Fe <sub>2</sub> O <sub>3</sub> | BaO  | K <sub>2</sub> O | TiO <sub>2</sub> | SrO  |
|-------------------|-----------------|-------|------------------|--------------------------------|--------------------------------|------|------------------|------------------|------|
| Constituent (wt%) | 56.23           | 36.73 | 5.38             | 0.57                           | 0.50                           | 0.31 | 0.15             | 0.06             | 0.06 |

Table 1. Composition and content of phosphogypsum.

| Treatment group | Soil-fertilizer ratios                        | Number of parallel treatments (pot) |
|-----------------|-----------------------------------------------|-------------------------------------|
| SS1             | 800 g limestone soil + 200 g SiO <sub>2</sub> | 6                                   |
| SS2             | 900 g limestone soil + 100 g SiO <sub>2</sub> | 6                                   |
| SP1             | 800 g limestone soil + 200 g phosphogypsum    | 6                                   |
| SP2             | 900 g limestone soil + 100 g phosphogypsum    | 6                                   |
| HS1             | 800 g yellow soil + 200 g SiO <sub>2</sub>    | 6                                   |
| HS2             | 900 g yellow soil + 100 g SiO <sub>2</sub>    | 6                                   |
| HP1             | 800 g yellow soil + 200 g phosphogypsum       | 6                                   |
| HP2             | 900 g yellow soil + 100 g phosphogypsum       | 6                                   |

**Table 2.** Proportion of soil and fertilizer in the experimental treatment group.

until the conclusion of the experiment on May 26, 2024. The setup of the pot system and the plant growth status at the end of the experiment are shown in Figure S1.

**Soil treatment and sampling**

For soil sampling, plants were carefully removed from the potting soil, and rhizosphere soil was collected using a sterilized toothpick and transferred into sterile sampling bags. Approximately 2 g of rhizosphere soil was collected per sample, with four replicates prepared for each treatment. An additional portion of rhizosphere soil was allocated to Ziploc™ bags (four replicates) for further analysis. Samples designated for microbial diversity assessment were stored at -80 °C. The soil contained in the Ziploc™ bags was dried at 45 °C, ground through a 60-mesh sieve, and subsequently used for determining soil physicochemical properties<sup>35</sup>.

**Plant sample processing**

Following soil sampling, plant rhizospheres were washed with clean water, and surface moisture was removed using tissues. Plant height was measured with a ruler, and the fresh mass of each plant was determined with an analytical electronic balance. Plant samples were oven-dried at 105 °C for 30 min, followed by overnight drying at 80 °C. Dry mass was then measured using an electronic analytical balance.

Subsequently, plant tissues were pulverized using an agate mortar. A 0.5 g sample of plant powder (in triplicate) was weighed into a 100-mL glass triangular flask. 10 mL of concentrated nitric acid and 5 mL of perchloric acid were added, followed by incubation at 100 °C for two hours on a rotating magnetic stirrer<sup>36</sup>. Additional nitric acid was added as needed to facilitate digestion. After digestion, the solution was transferred into a 50-mL glass cuvette. The triangular flask was rinsed repeatedly, and the final volume was adjusted to 25 mL before analyzing PTE concentrations (Pb, Cd, Cu, Zn, Cr, Mn) using atomic absorption spectroscopy (AAS). PTE concentrations were compared with China’s national food safety standards to evaluate the edibility.

**Soil physicochemical property determination**

Soil fertility was assessed through the determination of available metal and non-metallic elements using the glacial acetic acid leaching method<sup>37</sup>. Specifically, 5.7 mL of glacial acetic acid in double distilled water (ddH<sub>2</sub>O) was added to 1 L of extract solution. 2 g of soil sample (in triplicate) were weighed on an electronic balance, transferred into a 50-mL EP tube, and 20 mL of the glacial acetic acid extract was added using a pipette. The sample was shaken at 25 °C and 100 rpm for 24 h, followed by centrifugation at 9000 rpm, 25 °C, and 5 min to collect the supernatant. Concentrations of PTEs (Pb, Cd, Cu, Zn, Cr, Mn) and nutrient elements (K, Ca, Na, Mg, Fe) were determined using AAS. Total organic matter, total carbon, soluble total salt (according to Chinese National Standard NY/T 1121.16–2006) and soluble sulfate (according to Chinese National Standard GB/T 13025.8–2012) was analyzed gravimetrically. The contents of soil-related indices were calculated according to the correlation formula<sup>38</sup>. Additionally, soil oxalate content was determined by spectrophotometry<sup>39</sup>, carbonate content by acid-base titration<sup>40</sup>, and available phosphorus using the ammonium molybdate colorimetric method (according to Chinese National Standard GB 11893 – 1989). Soil pH (according to Chinese National Standard NY/T 1121.2–2006) and ionic conductivity (according to Chinese National Standard HJ 802–2016) were measured using a pH meter and conductivity meter, respectively. Statistical analysis was performed using one-way ANOVA followed by LSD post hoc tests (*P* = 0.05) using IBM SPSS Statistics 21 to identify significant differences.

Based on the concentrations of PTEs determined in the tested plant tissues and soil, the bioaccumulation factor (BAF) for each element in plant was calculated according to Formula (1)<sup>41</sup>:

$$BAF = \frac{\text{Concentration in plant tissues}}{\text{Concentration in soil}} \tag{1}$$

**Rhizosphere soil microbial diversity analysis**

*Bacterial and fungal DNA extraction and PCR amplification*

Rhizosphere soil samples were retrieved from storage at -80 °C. Bacterial DNA was extracted using the DNeasy PowerSoil Kit (Qiagen, Hilden, Germany) following the manufacturer’s instructions. DNA concentration and integrity were determined using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA) and agarose gel electrophoresis. The V3-V4 hypervariable region of the bacterial 16 S rRNA gene was amplified in a

25- $\mu$ L reaction system using primers 343 F (5'-TACGGRAGGCAGCAG-3') and 798R (5'-AGGGTATCTAATCCT-3'). PCR amplification was performed with the MagPure Soil DNA LQ Kit (Magan) using high-fidelity Taq polymerase. After extraction, purified DNA samples were stored at -20 °C.

In accordance with the instructions provided in the MagPure Soil DNA LQ Kit (Magan) manual, genomic DNA was extracted from plant rhizosphere soil samples. The concentration and purity were determined using a NanoDrop 2000 spectrophotometer (Thermo Fisher Scientific, USA) and agarose gel electrophoresis. The extracted DNA was stored at -20 °C and used as the template for polymerase chain reaction (PCR). The fungal internal transcribed spacer (ITS) genes using specific primers bearing barcodes and high-fidelity Takara Ex Taq enzyme. Universal primers ITS1F (5'-CTTGGTCATTTAGAGGAAGTAA-3') and ITS2 (5'-GCTGCGTTCTTCATCGATGC-3') were employed to amplify the variable region of the ITS gene ITS1, aiming to analyze fungal diversity<sup>42</sup>.

#### Library construction and sequencing

PCR amplicons were assessed by electrophoresis for quality assessment. Products were then purified via AMPure XP beads and served as a template for the second round of PCR amplification. A second purification step using magnetic beads was performed, and the final products were quantified using the Qubit assay. Concentration adjustment were made to prepare for downstream sequencing. DNA sequencing was conducted using the Illumina NovaSeq 6000 sequencing platform, generating 250 bp paired-end reads. Sequencing services were provided by OE Biotech Co., Ltd (Shanghai, China).

#### Bioinformatics analysis

The sequencing data were formatted as FASTQ files. The paired-end reads were preprocessed using the Cutadapt software to detect and remove residual adapters. After trimming, low-quality sequences were filtered, denoised, merged, and chimera sequences were identified and truncated using the DADA2 plugin in QIIME2 with default parameters<sup>43,44</sup>. This process yielded a feature table of representative reads and amplicon sequence variants (ASVs) abundance profile. Representative reads for each ASV were selected using the QIIME 2 package. All representative reads were annotated using the q2 feature classifier with default parameters and analyzed using the Silva database (Version 138; 16S/18S/ITS rDNA) or Unite.

Alpha and beta diversities were analyzed using QIIME 2. Alpha diversity, as measured by the Chao1 index and Shannon index, was used to assess microbial diversity in rhizosphere soil samples. For beta diversity analysis, the unweighted Unifrac distance matrix, calculated by R, was employed for unweighted Unifrac principal coordinate analysis (PCoA). Differential analysis was conducted using the R package with ANOVA, Kruskal-Wallis test, *T*-test, and Wilcoxon test. Species abundance profiles were analyzed using the LEfSe method. Both 16 S and ITS rRNA gene amplicon sequencing and subsequent analysis were provided by OE Biotech Co., Ltd (Shanghai, China).

The RDA method was used to investigate the correlations between environmental variables and microbial communities. Functional differences between bacterial and fungal species were identified using KEGG and COG.

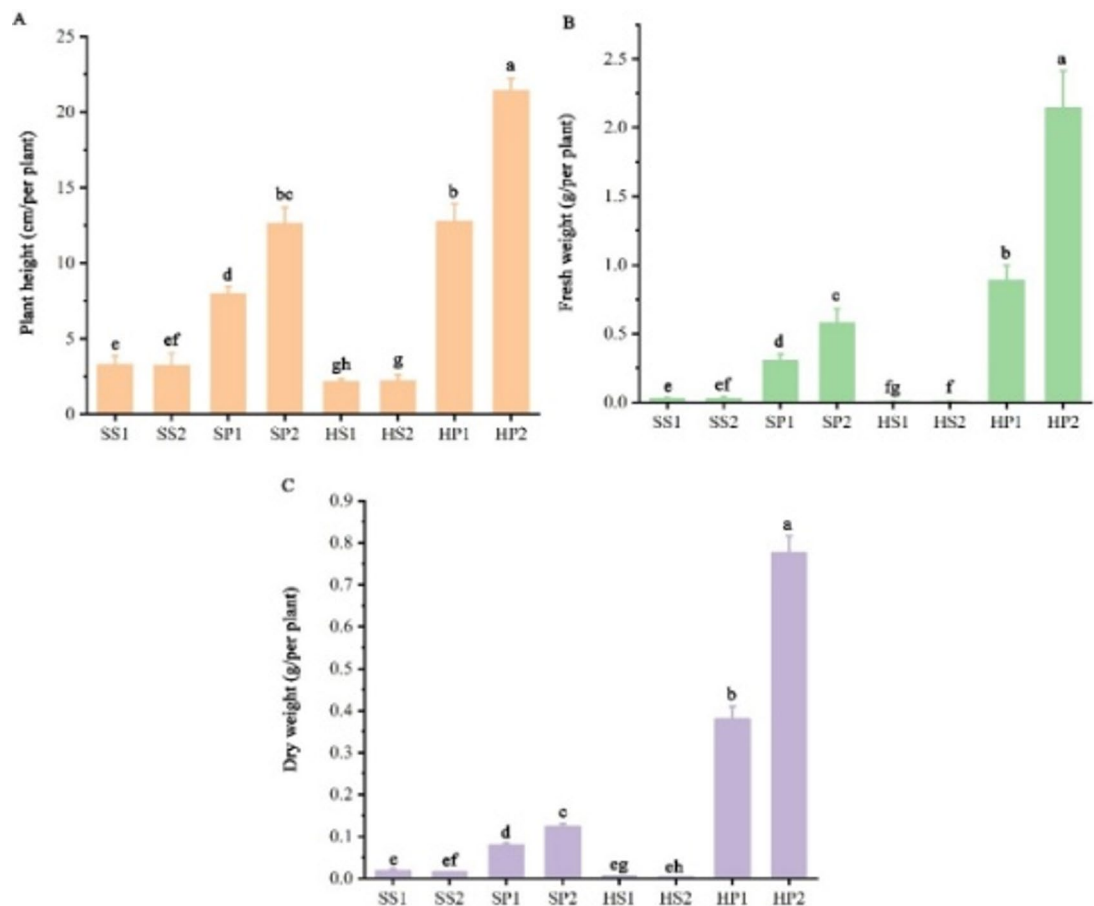
## Results and discussion

### Plant height, fresh and dry mass of *Amaranthus tricolor*

In the experiment, the growth performance of potted plants under different treatments demonstrates the effects of PG application (Figure S1). The results indicate that the growth of *A. tricolor* was significantly enhanced in both yellow soil and limestone soil treatments, following the PG addition, compared to that in the control group treated with SiO<sub>2</sub>. These findings demonstrate that PG incorporation facilitates plant growth. Specifically, plants in the 20% PG treatment exhibited greater growth compared to those in the 10% PG treatment. In contrast, control group plants showed stunted growth, indicating that both yellow and limestone soils possess inherently low fertility and cannot adequately support the plant growth optimally without supplementation. Phosphogypsum supplies essential nutrients for plant development, including phosphorus, calcium, sulfur and manganese. The results shown in Figure S1 further corroborate that PG addition enhances soil fertility and promotes the growth of *A. tricolor*. This evidence supports the potential of PG as a valuable soil amendment, particularly at high application rates.

Plant height, fresh mass, and dry mass of amaranth serve as key parameters for biological characterization in pot experiments. The detection results of the parameters of the pot-grown amaranth are shown in Fig. 1.

The results (Fig. 1) indicate that the originally barren yellow soil and limestone soil, after removing organic residues, drying, and grinding (as detailed in Materials and Methods), were even less conducive to the growth of *A. tricolor*. All control group plants exhibited extremely poor growth, with plant heights of only about 2–3 cm, and no statistically significant differences in plant height between the control groups (without PG addition). It is noteworthy that the pot experiment was conducted in an open-air ventilated greenhouse during a period (March–May 2024) of unusually high temperatures in Nanjing. Under these conditions, the plant height, dry weight, and fresh weight of *A. tricolor* in the high dosage PG treatment group significantly increased (Fig. 1). This indicates that PG not only enhances soil fertility but also improves soil water retention capacity and plant stress tolerance. Following the addition of 10% PG, the average plant heights of *A. tricolor* in yellow soil and limestone soil reached 12.7 cm and 7.9 cm, respectively. When PG addition reached 20%, the average plant heights further increased to 21.4 cm in yellow soil and 12.6 cm in limestone soil. Additionally, plants grown in yellow soil exhibited greater fresh and dry weights than those in limestone soil, whereas control group showed lower fresh and dry weights compared to those in limestone soil. This discrepancy may be attributed to inherent differences in soil fertility between the two soil types, with limestone soil exhibiting higher initial fertility than yellow soil.



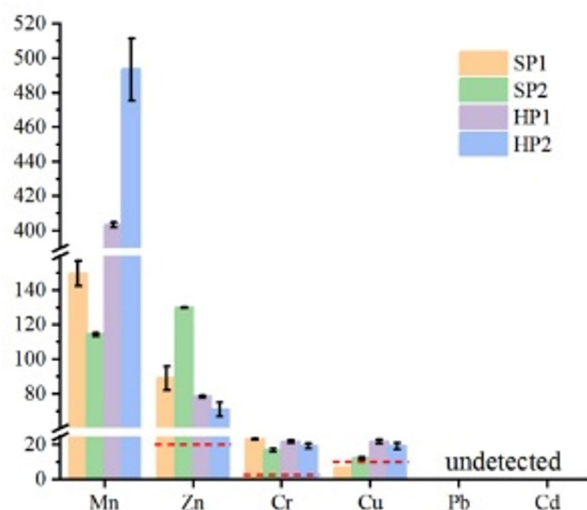
**Fig. 1.** Determination of biological indexes of *Amaranthus tricolor* (A) Plant height (B) Fresh mass (C) Dry mass (letters indicate significant differences, using One way ANOVA, LSD,  $P = 0.05$ ).

The addition of PG significantly altered the growth conditions of *A. tricolor*, demonstrating its effectiveness in ameliorating barren soil and suggesting considerable potential for agricultural applications.

Importantly, the PTE content in *A. tricolor* must be assessed to ensure its safety for consumption. As shown in Fig. 2, except for Pb and Cd, which were below detection limits, the concentrations of Zn ( $\leq 20$  mg/kg, according to GB 5009.14–2017), Cr ( $\leq 0.5$  mg/kg, according to GB 2762–2022), and Cu ( $\leq 10$  mg/kg, according to GB 5009.13–2017) in *A. tricolor* exceeded acceptable limits. Furthermore, Chinese regulations have not established specific thresholds for Mn content in food products. These findings suggest that *A. tricolor* cultivated in soil treated with high dosage of PG may pose health risks due to elevated PTE concentrations. It is noteworthy that no phytotoxic effects were observed phenotypically; however, whether phytotoxicity occurs at the cellular and molecular levels in *A. tricolor* remains to be investigated. Additionally, considering that the excessive PTEs accumulation in plants may be attributed to intensive transpiration under high-temperature conditions during the plant growth, more comprehensive ecotoxicological assessments are warranted. The obtained *A. tricolor* was found to be more suitable for applications such as biomass power generation, bioethanol production, and biochar preparation. In light of larger-scale field experiments, the environmental safety risks associated with high dosage phosphogypsum application require further investigation. Meanwhile, the feasibility of directly cultivating non-edible plants and ornamental species under such conditions warrants systematic investigation to comprehensively evaluate their practical implementation potential.

### Soil physicochemical properties

The efficacy of PG soilization depends on its capacity to enhance soil physicochemical attributes and improve fertility without causing significant contamination. Therefore, the characterization of the physicochemical properties of the experimental soils is a critical aspect of this study. As detailed in Table 3, the results of soil analyses demonstrate that PG addition significantly increased concentrations of available P, total salts, calcium, sodium, copper, chromium, manganese, and organic carbon in both yellow and limestone soils. These effects were notably influenced by the PG application rate. In contrast, electrical conductivity and the concentrations of available potassium, magnesium, zinc, and iron increased in limestone soil but showed a decreasing trend in yellow soil. Additionally, the pH of limestone soil decreased significantly after PG addition, while the pH of yellow soil increased notably.



**Fig. 2.** Determination of the PTE content in *Amaranthus tricolor*, the red dash represents the acceptable limits of corresponding PTEs.

| Soil index                      | Limestone soil    |                   |                  |                   | Yellow soil     |                   |                  |                   |
|---------------------------------|-------------------|-------------------|------------------|-------------------|-----------------|-------------------|------------------|-------------------|
|                                 | SS1               | SS2               | SP1              | SP2               | HS1             | HS2               | HP1              | HP2               |
| A. P (mg/kg)                    | 4.43 ± 0.07 c     | 4.78 ± 0.13 c     | 42.99 ± 0.03 a   | 39.40 ± 1.12 b    | -               | -                 | 2.01 ± 0.16 b    | 2.83 ± 0.06 a     |
| A. K (mg/kg)                    | 43.27 ± 0.23 c    | 38.35 ± 0.16 d    | 47.80 ± 0.54 b   | 66.44 ± 2.29 a    | 54.67 ± 1.53 a  | 45.37 ± 0.61 b    | 29.33 ± 0.23 d   | 34.40 ± 0.38 c    |
| T. C %                          | 4.33 ± 0.22 c     | 4.21 ± 0.00 cd    | 7.87 ± 0.01 a    | 6.09 ± 0.26 b     | 5.71 ± 0.06 c   | 4.12 ± 0.02 d     | 6.72 ± 0.20 b    | 6.91 ± 0.08 a     |
| T. O %                          | 7.47 ± 0.22 c     | 7.26 ± 0.00 cd    | 13.56 ± 0.01 a   | 10.49 ± 0.26 b    | 9.84 ± 0.06 c   | 7.11 ± 0.02 d     | 11.58 ± 0.20 b   | 11.92 ± 0.08 a    |
| CaCO <sub>3</sub> %             | 0.65 ± 0.00 b     | 0.74 ± 0.00 a     | 0.70 ± 0.06 ab   | 0.70 ± 0.06 ab    | 0.29 ± 0.00 a   | 0.34 ± 0.06 a     | 0.16 ± 0.06 c    | 0.21 ± 0.00 bc    |
| T. S (g/kg)                     | 3.30 ± 0.00 b     | 3.60 ± 2.69 b     | 33.70 ± 1.13 a   | 35.05 ± 0.92 a    | 2.80 ± 0.00 bc  | 4.10 ± 1.27 b     | 36.65 ± 1.77 a   | 36.60 ± 2.12 a    |
| pH                              | 7.18 ± 0.15 b     | 7.45 ± 0.05 a     | 7.00 ± 0.02 c    | 6.93 ± 0.06 d     | 5.40 ± 0.05 c   | 5.69 ± 0.08 b     | 5.91 ± 0.15 a    | 6.03 ± 0.09 a     |
| E. C (μs/cm)                    | 140.20 ± 1.41 a   | 135.80 ± 4.24 a   | 2.17 ± 0.01 bc   | 2.28 ± 0.01 b     | 66.65 ± 2.47 a  | 60.80 ± 0.14 b    | 2.13 ± 0.05 cd   | 2.17 ± 0.02 c     |
| SO <sub>4</sub> <sup>2-</sup> % | -                 | -                 | 1.98 ± 0.05 a    | 1.98 ± 0.15 a     | 0.15 ± 0.07 b   | 0.06 ± 0.02 b     | 1.99 ± 0.11 a    | 1.99 ± 0.07 a     |
| Oxalate (μg/kg)                 | 23.11 ± 0.00 a    | 16.81 ± 1.20 b    | 18.68 ± 0.48 b   | 25.00 ± 2.17 a    | 22.43 ± 0.96 c  | 21.92 ± 0.72 c    | 26.35 ± 0.72 b   | 29.07 ± 0.72 a    |
| A. Ca (mg/kg)                   | 1766.30 ± 32.93 c | 1561.29 ± 21.60 d | 3590.09 ± 5.67 a | 3476.33 ± 18.63 b | 327.87 ± 8.10 b | 251.01 ± 63.45 bc | 3138.04 ± 4.66 a | 3102.41 ± 17.82 a |
| A. Na (mg/kg)                   | 37.34 ± 1.07 d    | 39.12 ± 0.30 c    | 56.21 ± 0.00 a   | 50.15 ± 3.04 b    | 46.96 ± 2.77 c  | 46.50 ± 0.00 c    | 56.00 ± 0.29 b   | 73.41 ± 0.84 a    |
| A. Mg (mg/kg)                   | 91.84 ± 0.00 c    | 79.86 ± 0.48 d    | 119.51 ± 4.16 b  | 137.03 ± 1.52 a   | 129.89 ± 6.56 a | 102.83 ± 5.00 b   | 98.45 ± 3.77 bd  | 101.15 ± 0.88 bc  |
| A. Cu (mg/kg)                   | 0.26 ± 0.00 c     | 0.27 ± 0.01 c     | 0.42 ± 0.00 b    | 0.60 ± 0.01 a     | 0.32 ± 0.00 c   | 0.29 ± 0.01 d     | 0.60 ± 0.00 b    | 0.73 ± 0.00 a     |
| A. Zn (mg/kg)                   | 0.03 ± 0.00 d     | 0.08 ± 0.00 c     | 0.36 ± 0.01 b    | 0.71 ± 0.02 a     | 0.40 ± 0.00 a   | 0.33 ± 0.01 b     | 0.15 ± 0.02 d    | 0.27 ± 0.00 c     |
| A. Cr (mg/kg)                   | 0.16 ± 0.03 d     | 0.79 ± 0.00 c     | 1.43 ± 0.00 b    | 1.68 ± 0.07 a     | -               | -                 | 0.22 ± 0.00 b    | 0.42 ± 0.00 a     |
| A. Mn (mg/kg)                   | 90.60 ± 1.45 b    | 73.84 ± 3.43 c    | 90.63 ± 1.82 b   | 121.80 ± 5.25 a   | 18.20 ± 0.34 c  | 15.10 ± 0.90 d    | 27.49 ± 1.17 b   | 36.91 ± 3.33 a    |
| A. Fe (mg/kg)                   | 0.61 ± 0.03 d     | 0.83 ± 0.03 c     | 1.48 ± 0.00 b    | 1.61 ± 0.00 a     | 0.81 ± 0.00 ab  | 1.08 ± 0.09 a     | 0.68 ± 0.06 bc   | 0.74 ± 0.21 b     |
| A. Pb                           | -                 | -                 | -                | -                 | -               | -                 | -                | -                 |
| A. Cd                           | -                 | -                 | -                | -                 | -               | -                 | -                | -                 |

**Table 3.** The soil physicochemical indicators. -: undetected; A. X: effective state content (X = K, Pb, Cd, Zn, etc.); T. C: total carbon; T. O: total organic matter; T. S: total water-soluble salts; E. C: conductivity; CaCO<sub>3</sub> %: calcium carbonate content of soil.

PG consists primarily of calcium sulfate (pH 2–5) along with various PTEs, soluble salts, and insoluble mineral constituents. The soilization process facilitates the migration of free elements and insoluble minerals into the soil, resulting in increased concentrations of available P, total salts, calcium, sodium, copper, chromium, manganese, total carbon, organic matter, and oxalate (Table 3). The greater biomass of *A. tricolor* observed in yellow soil (Fig. 1) may be attributed to its more efficient uptake and utilization of soil nutrients compared to limestone soil. This is evidenced by the lower electrical conductivity and reduced concentrations of available potassium, magnesium, zinc, and iron in yellow soil compared to the control group. Conversely, the lower biomass of *A. tricolor* in limestone soil was associated with higher soil electrical conductivity and available nutrient concentrations compared to the control group. Given that limestone soil is weakly alkaline, its interaction with PG likely induces

acid-base neutralization, leading to the observed pH reduction after PG addition. The increased pH of yellow soil may be related to the fixation of Fe or Al by PG or the production of alkaline substances by the soil microbial community<sup>45,46</sup>. Regarding PTE concentrations in soil, according to the GB 15,618 – 2018, the risk thresholds for Cu, Zn, and Cr are 100, 250, and 200 mg/kg, respectively, when the pH ranges from 6.5 to 7.5. For a pH range of 5.5 to 6.5, the corresponding risk thresholds are 50, 200, and 150 mg/kg. The results indicate that in both yellow soil and limestone soil, the contents of Cu, Zn, and Cr after the addition of PG are significantly lower than these thresholds. This suggests that high dosage incorporation of PG soilization following *A. tricolor* cultivation does not lead to the accumulation of PTEs in the soil but rather enhances the nutrient content in the soil. *A. tricolor* secretes oxalic acid and other organic acids through its roots, which weather minerals and lead to increased soil oxalate and organic matter/organic carbon content<sup>47</sup>. Compared with the control group, the addition of phosphogypsum did not lead to a significant increase in soil PTEs, which also indicates that high application rates of phosphogypsum did not induce PTE stress in *A. tricolor*. The significant enhancement in soil organic matter and *A. tricolor* biomass suggests that high dosage PG incorporation acts as a soil carbon sink. Compared with other studies on the soil application of PG in Karst region, Yin et al.<sup>48,49</sup> conducted a survey on soil sulfur fertility in Guizhou as early as 1991–1992, revealing that approximately one-third of the soils in Guizhou were sulfur-deficient or potentially sulfur-deficient, while PG application could supply sufficient sulfur to support high and stable yields of crops such as rice, maize, wheat, and rapeseed. Research on PG as an amendment for acidic soils such as yellow soil showed that soil pH increased by 0.2–0.5. Calcium, magnesium, sulfur, phosphorus, and silicon introduced by PG application are excellent plant nutrients for farmland crops, improving soil structure and promoting broomcorn growth<sup>50</sup>. Although existing studies involve different plant species and PG application rates, the same consensus is that PG used for amending karst soils has demonstrated considerable potential in enhancing soil water retention and nutrient content, as well as effectively controlling the risk of heavy metal pollution<sup>50–52</sup>. However, further studies are required to evaluate the long-term effects of high dosage PG incorporation.

Bioaccumulation factor (BAF) can reflect the capacity of *A. tricolor* to accumulate elements and assist in evaluating its remediation effect on soils subjected to high dosage PG incorporation. The BAF values of *A. tricolor* in each experimental group are presented in Table 4. The results indicate that, except for the low BAF value of Mn in plants grown in limestone soil, *A. tricolor* in all groups exhibited relatively high BAF values for the measured elements. On one hand, this phenomenon may be attributed to enhanced transpiration caused by high-temperature conditions; on the other hand, it suggests that *A. tricolor* possesses considerable potential for accumulating PTEs in soils amended with high dosage of PG. Although this trait makes the plant unsuitable for food or feed purposes, it demonstrates their excellent capability for remediating PTE contaminant soil. This also explains why, despite the high PG incorporation, the concentrations of various PTEs in the soil remained well below risk thresholds.

The aforementioned studies indicate that high dosage incorporation of PG soilization not only improves soil physicochemical properties and fertility, but also prevents significant soil contamination (Table 2), while stimulating plant growth (Fig. 1 and Figure S1). Regarding the potential risk of radioactive contamination from phosphogypsum<sup>53</sup>, a recent investigation and evaluation conducted by the Guizhou Provincial Radiation Environment Monitoring Station indicates that the radioactivity level of phosphate ores in the main phosphate mining areas of Guizhou is comparable to the background radioactivity level of local soils. Moreover, the radioactivity level of phosphogypsum and its derived products is generally lower than the soil background level in Guizhou<sup>54</sup>. Therefore, the comprehensive utilization of phosphogypsum in Guizhou does not pose additional radiation contamination to the environment or individuals. These findings suggest that the utilization of high dosage incorporation of PG soilization in the karst region of Guizhou represents a promising approach. However, the potential risks related to fluoride hazards associated with PG remain to be thoroughly investigated, as it may become substantially elevated under specific environmental conditions<sup>55</sup>.

### Structural analysis and functional prediction of soil microbial communities

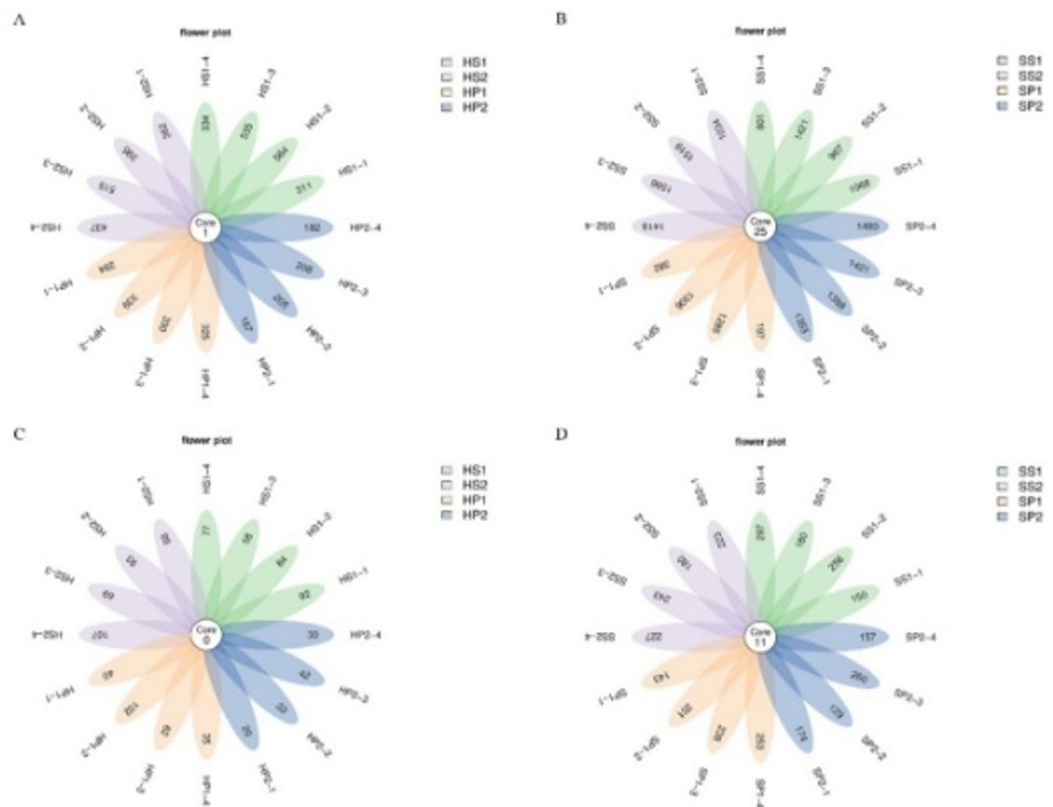
The composition of microbial communities serves as a crucial indicator for assessing soil health. The structural analysis of the rhizosphere soil microbial community in *A. tricolor* grown in PG pots is detailed as follows.

#### ASV classification petal diagram and community structure histogram

Following microbial diversity sequencing, the obtained ASVs from the samples were classified. The results are presented in Fig. 3. As shown in Fig. 3A, a single shared ASV was identified between the yellow soil treatment and control groups. The incorporation of 20% PG was associated with a significant reduction in bacterial ASVs in yellow soil. In contrast, the total number of bacterial ASVs in the limestone soil treatment and control groups remained stable at 25, with negligible effects of PG addition (Fig. 3B). The addition of PG resulted in a reduction in the number of common ASV in yellow soil fungi and a decrease in the overall number of fungal

| Treatment group | Mn    | Zn     | Cr    | Cu    |
|-----------------|-------|--------|-------|-------|
| SP1             | 1.65  | 247.39 | 16.34 | 16.29 |
| SP2             | 0.94  | 183.11 | 10.09 | 20.47 |
| HP1             | 14.68 | 522.93 | 99.27 | 36.60 |
| HP2             | 13.37 | 263.19 | 46.29 | 26.66 |

**Table 4.** Bioaccumulation factors (BAF) of the *Amaranthus tricolor*.



**Fig. 3.** The amplicon sequencing variant classification petal map (A) Yellow soil bacteria (B) Limestone soil bacteria (C) Yellow soil fungi (D) Limestone soil fungi.

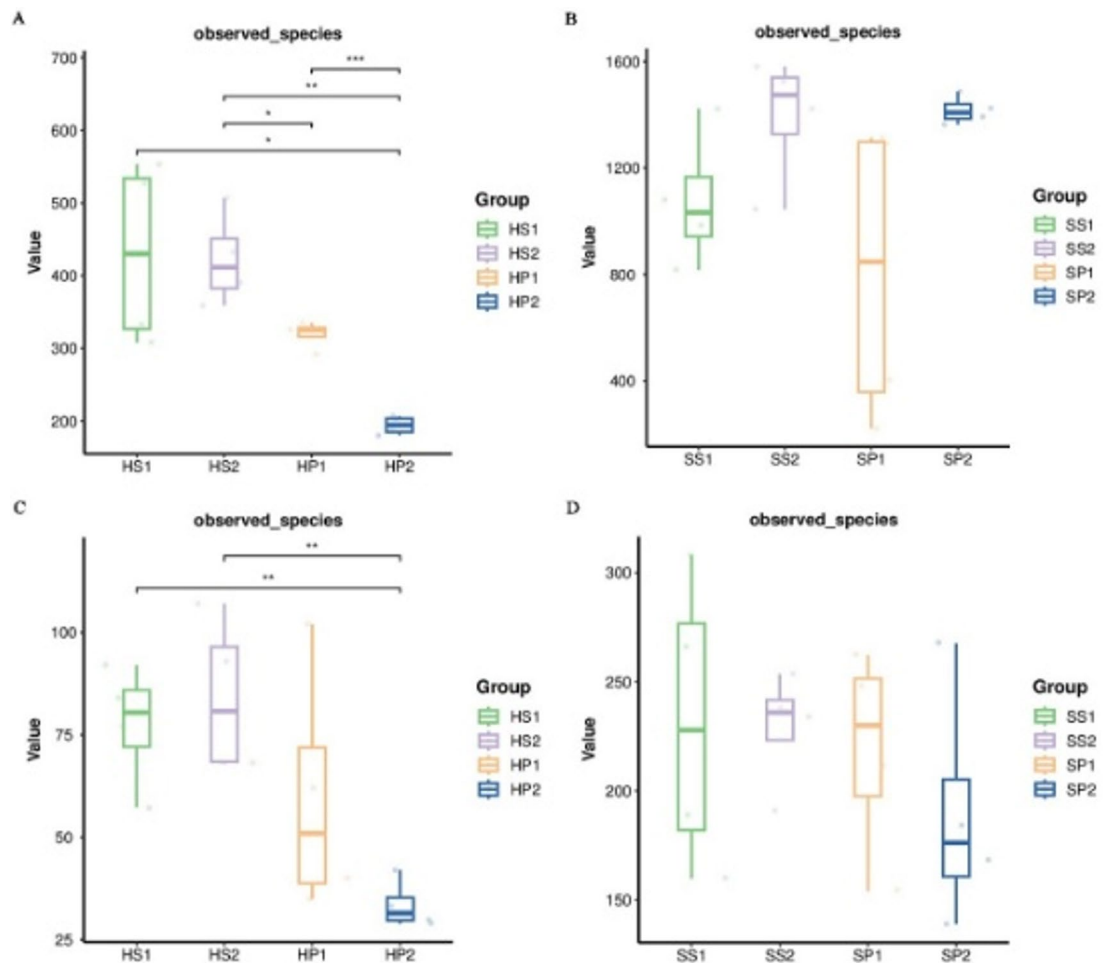
ASV (Fig. 3C). Notably, the introduction of PG resulted in a decline in shared fungal ASVs in limestone soil and had minimal effect on the overall fungal ASV count (Fig. 3C). In limestone soil, the application of PG led to the detection of 11 unique fungal ASVs, with minimal impact on the total fungal ASV count (Fig. 3D). Collectively, these results indicate that PG addition tended to reduce bacterial and fungal abundance in yellow soil. Soil pH and nutrient availability are key factors shaping microbial community structure and composition. The research found that yellow soil was weakly acidic ( $\text{pH } 5.79 \pm 0.05$ ), and limestone soil was near-neutral ( $\text{pH } 7.21 \pm 0.11$ ). Given its richer nutrient profile (Table S1), near-neutral pH and higher nutrient availability, limestone soil is more favorable for microbial community proliferation. The petal map results further demonstrate that limestone soil has a higher diversity of bacterial and fungal species (Figure S2). The differences in the effects of PG on microbial communities between limestone soil and yellow soil may be related to the differences in soil pH buffering capacity and physicochemical properties.

#### Alpha diversity analysis

Analysis of microbial community structure offers insights into compositional uniformity and diversity within samples. Alpha diversity results for the soil sample are presented in Fig. 4, Figure S3, and Table S2. As shown in Figure S3, the species discovery curve tends to flatten with increasing sequencing depth, indicating that sequencing depth is adequate and the observed species richness is appropriate for subsequent diversity analysis. Alpha diversity boxplots are presented in Fig. 4, which depict the distribution of diversity indices within each group and facilitate the comparison of significant intergroup differences. As demonstrated in Fig. 4A, PG incorporation significantly reduced bacterial diversity in yellow soil, with notable disparities in observed species between treatment groups. In contrast, as shown in Fig. 4B, PG exhibited minimal impact on bacterial diversity in limestone soil, and the observed difference was not statistically significant. The effects of PG on fungal communities are illustrated in Fig. 4C and D. As shown in Fig. 4C, the addition of 20% PG led to a noticeable reduction in the observable fungal population in yellow soil. In comparison, as depicted in Fig. 4D, the influence of PG on fungal composition in limestone soil was relatively minor. These findings are consistent with the alpha diversity boxplot results in Table S2, reinforcing treatment-specific patterns in microbial diversity.

#### Beta diversity analysis

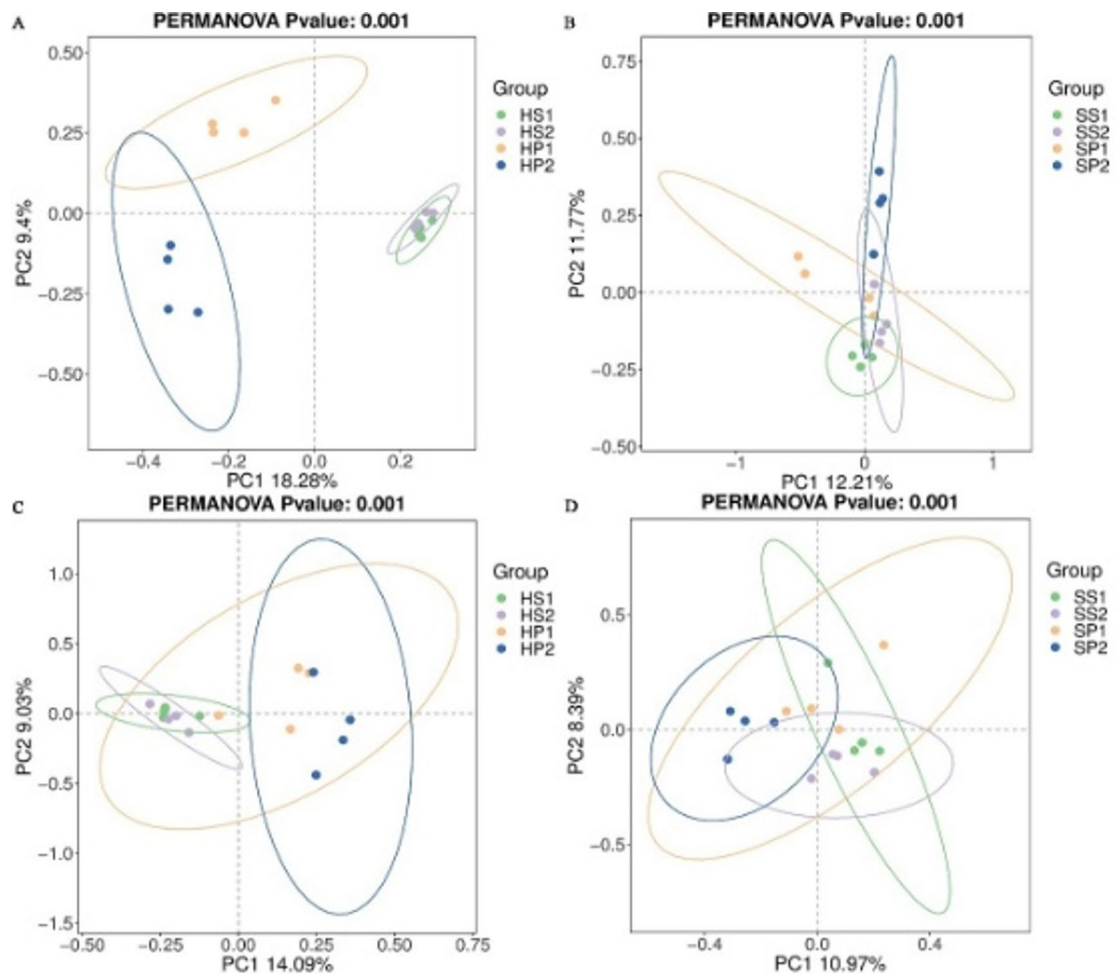
Beta diversity analysis evaluates differences in microbial communities between sample groups by examining habitat-specific compositional variation, incorporating both phylogenetic relationships among ASVs and community structural differences. Beta diversity results are presented in Fig. 5, which employs principal coordinate analysis (PCoA) to visualize these differences. As demonstrated in Fig. 5A, the PCoA plot reveals



**Fig. 4.** The alpha diversity correlation boxplot.

clustering of samples by color, with distinct separations along PC1 and PC2 axes. These findings suggest that the incorporation of PG, particularly at varying addition levels, significantly influences the structure and composition of bacterial communities in yellow soil. Notably, the clustering of control group samples indicates that variations in SiO<sub>2</sub> addition have negligible impact on the bacterial community. In contrast, as illustrated in Fig. 5B, the differences between sample groups are less pronounced, with minimal dispersion along PC2 axis, implying that the addition of PG and its dosage adjustments exert limited influence on bacterial community composition in limestone soil. Furthermore, as shown in Fig. 5C, fungal community composition in yellow soil is affected by PG incorporation, as evidenced by distinct separation of sample points between the PG-treated and control groups along PC1 axis. However, this effect appears modest and not directly proportional to PG dosage. As depicted in Fig. 5D, the dispersion patterns of sample points along both PC1 and PC2 axes suggests that PG incorporation has a relatively minor impact on fungal community structure.

These beta diversity results highlight differential responses of bacterial and fungal communities to PG addition across soil types. Specific variations in microbial community composition at the phylum level are further elaborated in Fig. 6. The findings in Fig. 6A indicate that PG addition notably alters bacterial community composition in yellow soil compared to the control group, with a significant increase in the relative abundance of Bacteroidota and Firmicutes, accompanied by a marked decrease in Proteobacteria and Acidobacteria. Moreover, increasing PG dosage correlates with declining proportions of Proteobacteria and Bacteroidota, while Acidobacteria and Campilobacterota increased correspondingly. In contrast, as demonstrated in Fig. 6B, PG incorporation does not induce substantial changes in bacterial communities in limestone soil, though slight elevations in Patescibacteria and Proteobacteria abundance are observed. The results presented in Fig. 6C reveal that PG addition to yellow soil increased the relative abundances of Basidiomycota and Glomeromycota while reducing Chytridiomycota. Notably, variations in PG dosage exhibit minimal impact on fungal composition in yellow soil. As illustrated in Fig. 6D, the incorporation of PG into limestone soil results in a significant increase in Basidiomycota abundance and reduced Ascomycota relative abundance. Furthermore, the degree of Ascomycota reduction is negatively correlated with increasing PG dosage. In summary, PG addition exerts a pronounced effect on bacterial communities in yellow soil, while its impact on fungal communities remains relatively limited. Conversely, the influence of PG on limestone soil microbial communities is more evident for



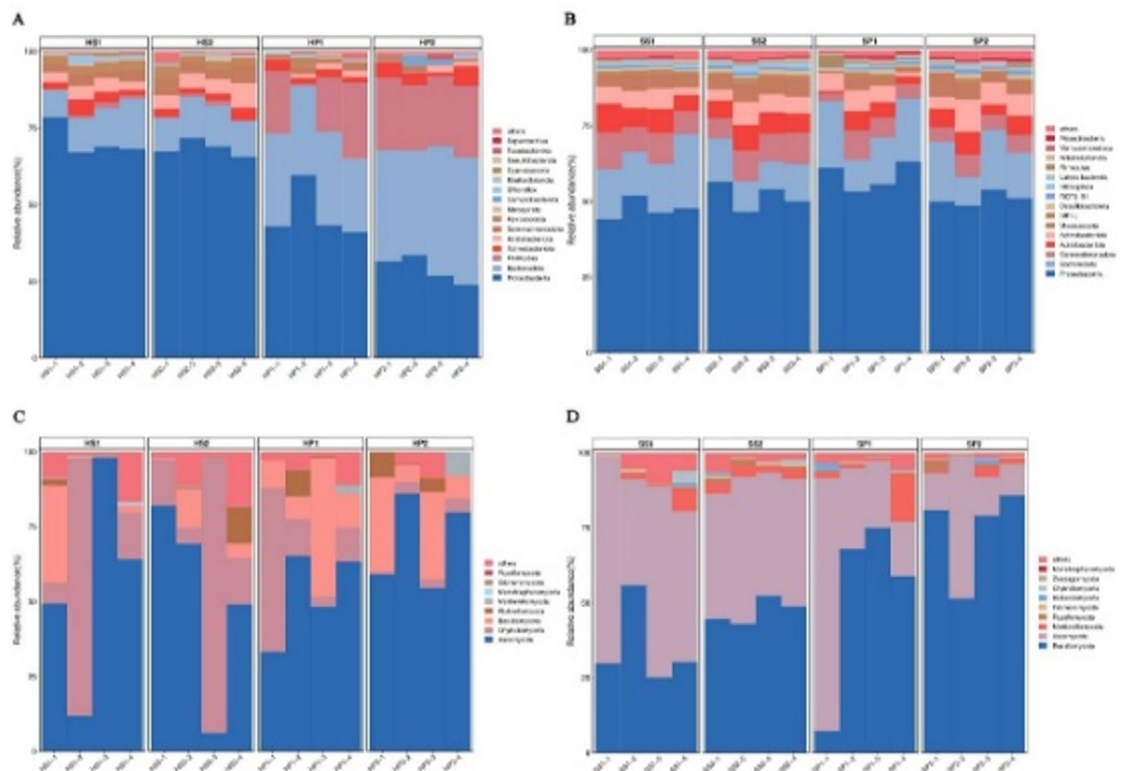
**Fig. 5.** The PCoA analysis in the comparison of beta diversity (A) Yellow soil bacteria (B) Limestone soil bacteria (C) Yellow soil fungi (D) Limestone soil fungi (binary-Jaccard distance algorithm).

fungi than for bacteria. Notably, the magnitude of these effects is positively correlated with PG dosage, though distinct response patterns are observed between soil types and microbial domains.

Based on the aforementioned analysis of microbial community structure, it can be observed that high dosage PG incorporation significantly affects the microbial community structure of yellow soil. Specifically, bacterial and fungal ASV counts decreased notably, bacterial diversity has significantly declined, and community composition underwent considerable changes. Furthermore, the fungal community was influenced to some extent. On the other hand, the microbial community structure and diversity of limestone soil remained largely unchanged following PG addition. These findings indicate that the effects of high dosage of phosphogypsum on yellow soil are more pronounced compared to limestone soil. However, the long-term changes in these impacts warrant further investigation.

#### *Analysis of the correlation between environmental factors and microbial diversity*

The specific characteristics of microenvironments play a critical role in shaping microbial community structures, depending on the prevailing environmental conditions. Canonical correspondence analysis (CCA) was employed to elucidate relationships between environmental factors and microbial communities. As depicted in Fig. 7, in yellow soil, positive correlations were observed between PG addition and various chemical and physical parameters, including Ca, Na, Cu,  $\text{SO}_4^{2-}$ , T.C, T.O, P, pH, Cr, T.S, Mn, and oxalate (Fig. 7A). Conversely, negative correlations were noted with electrical conductivity, Fe, Mg, K, and Zn. In limestone soil, a wide range of elements and parameters showed positive correlations with microbial communities in the PG-treated group, including Ca, Na, Cu,  $\text{SO}_4^{2-}$ , T.C, T.O, P, pH, Cr, T.S, Mn, oxalate, K, Zn, Fe, and Mg (Fig. 7B). However, pH and electrical conductivity exhibited negative correlations. Regarding fungal communities in yellow soil, positive correlations were observed with pH, Ca, P, Cu, Cr, Mn, Na,  $\text{SO}_4^{2-}$ , T.C, and T.O, whereas negative correlations were found with K, Zn, Mg, Fe, and electrical conductivity (Fig. 7C). In limestone soil, fungal communities were positively correlated with Ca, P, Cu, Cr, Mn, oxalate, Na,  $\text{SO}_4^{2-}$ , T.C, T.O, K, Zn, Mg, and Fe, but negative correlations with pH and conductivity (Fig. 7D). Overall, comparing the soil physicochemical indicators before and after the application of PG, the addition of high dosage phosphogypsum resulted in positive correlations between soil microbial communities and factors such as oxalate, P, Cu, Mn,  $\text{SO}_4^{2-}$ , T.S, Ca, Na, Cr, T.C, and



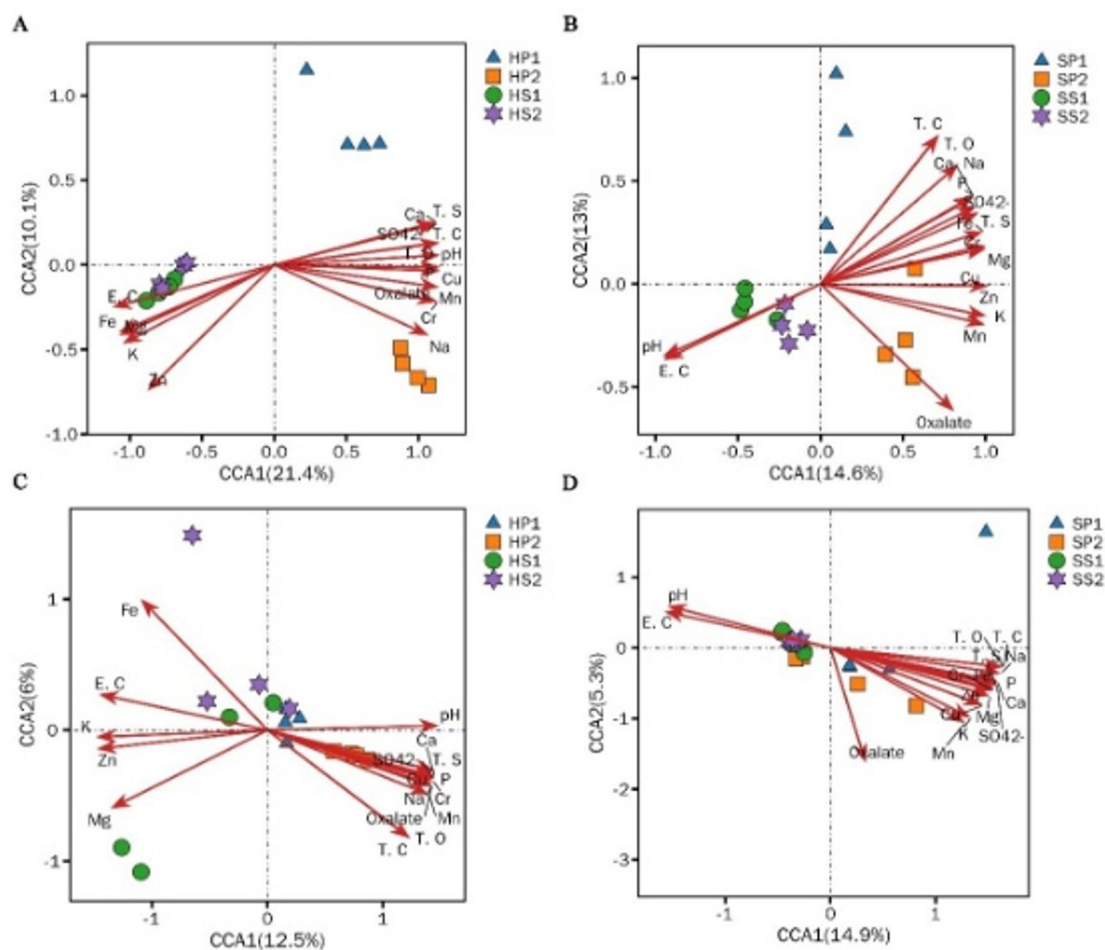
**Fig. 6.** The microbial community composition based on the classification level of phylum (A) Yellow soil bacteria (B) Limestone soil bacteria (C) Yellow soil fungi (D) Limestone soil fungi.

T.O. Conversely, electrical conductivity exhibited a negative correlation. Notably, oxalate, T.C, and T.O were significantly influenced by plant growth, while changes in other factors were predominantly attributed to PG addition.

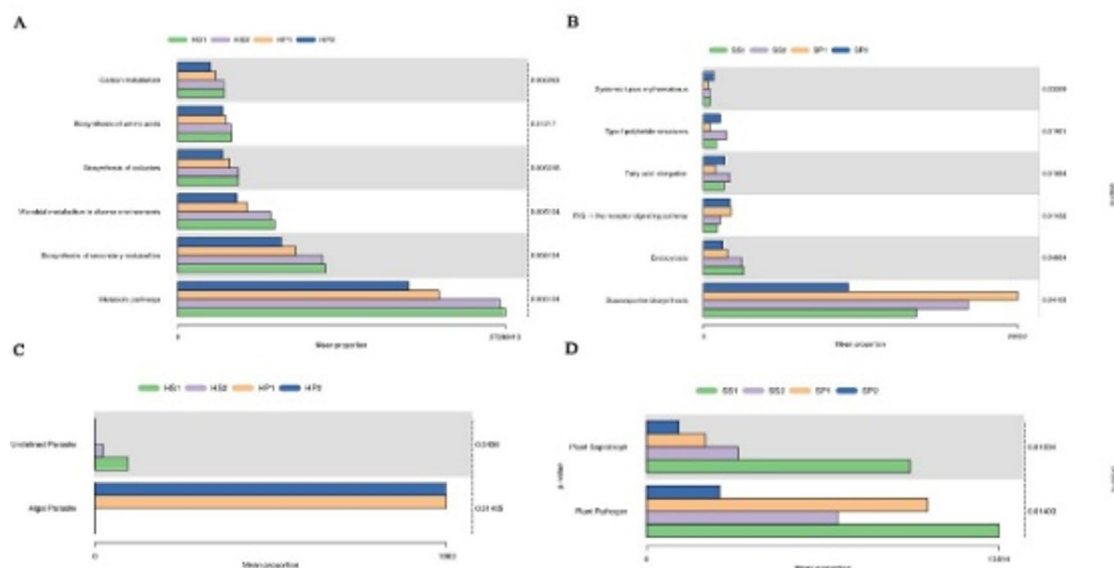
#### Prediction of microbial community function

KEGG functional prediction was conducted to identify functional differences between samples and groups by statistically determining the composition of known microbial gene functions. The results are presented in Fig. 8. As shown in Fig. 8A, PG incorporation into yellow soil significantly altered pathways related to carbon metabolism, amino acid biosynthesis, cofactor biosynthesis, microbial metabolism in diverse environments, and secondary and metabolic pathways. In limestone soil, PG addition was associated with increased representation of systemic lupus erythematosus, RIG-I-like receptor signaling, and related pathways, while type I polyketide structure, fatty acid elongation, and endocytosis metabolism decreased (Fig. 8B). Following PG addition to yellow soil, the relative abundance of algal parasites increased, accompanied by a significant reduction in undefined parasites (Fig. 8C). In limestone soil, plant saprotrophs and plant pathogenic fungi exhibited notable decreases (Fig. 8D). These findings suggest that PG incorporation impedes microbial proliferation and suppresses the growth of detrimental fungi. The observed increase in algal parasites may be associated with enhanced soil algal populations.

As a bulk solid mineral waste, environmental accumulation of PG can lead to significant acidification and PTE contamination, posing risks to the surrounding ecosystems<sup>56</sup>. However, high dosage incorporation of PG into karst mountainous soils in Guizhou presents promising opportunities for addressing challenges related to PG management—such as pollution control, resource utilization, and soil improvement—while also supporting the development of new cultivated soils in karst regions. This approach may help mitigate local ecological and economic issues, such as thin soil layers, poor soil quality, and fragile ecosystems<sup>57,58</sup>. The findings underscore that high dosage PG incorporation can enhance soil nutrients, optimize microbial community structures, and promote plant growth. However, it is important to note that this practice may lead to excessive accumulation of PTEs in plants, potentially posing risks to food safety, as well as potential radioactive contamination. Notably, in the limestone soil and yellow soil used in this study, no soil contamination resulting from high dosage PG incorporation was observed. However, the effects on other types of soils still require further evaluation. Based on these findings, a safe utilization model for high dosage PG soilization was proposed for karst mountainous regions of Guizhou. PG can be applied to the land through plowing, particularly in areas with thin soil layers, where the application rate can be increased to several tens of tons per hectare. Subsequently, plants with high accumulation capacity for PTEs should be selected for soil remediation. These non-food plants can extract PTEs from soil, mitigate PTEs' risk, enhance soil productivity, and improve the soil that has already been supplemented



**Fig. 7.** Analysis of the correlation between environmental factors and microbial communities.



**Fig. 8.** The differences analysis of microbial metabolic pathways in samples (A) Yellow soil bacteria (B) Limestone soil bacteria (C) Yellow soil fungi (D) Limestone soil fungi (Kruskal-Wallis algorithm, significantly different TOP 15).

with high dosage PG incorporation within a specified timeframe. Following this soil improvement phase, it is anticipated that the soil will be suitable for agricultural use. Nevertheless, comprehensive assessments of soil physicochemical properties and PTE concentrations remain necessary.

## Conclusions

The high dosage PG incorporation into soils from karst area has shown potential to improve soil fertility, enhance organic matter accumulation, optimize soil structure, regulate microbial community structures, and stimulate plant growth. This approach represents an effective and feasible approach for addressing the large-scale accumulation of phosphogypsum waste. However, this practice also poses potential ecological risks, particularly due to elevated PTE concentrations in cultivated plants. To mitigate these risks while maximizing the benefits of PG utilization, a synergistic approach combining high dosage PG incorporation with targeted selection of plants and microorganisms is proposed. This strategy involves an initial phase of PTE extraction using hyperaccumulator plant species and barrier factor reduction over a defined period, creating favorable conditions for subsequent agricultural use of the soil. By implementing this integrated method, the environmental and agronomic benefits of PG use can be achieved while minimizing potential ecological risks. This study mainly considered issues related to PTEs, alterations in soil physicochemical properties, and microbial community structure; however, further assessment of radioactive elements and fluoride pollution caused by long-term use is still needed.

## Data availability

The datasets used and analyzed during the current study are available on reasonable request.

Received: 30 September 2025; Accepted: 23 January 2026

Published online: 25 January 2026

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## Acknowledgements

We sincerely thank the editor's work and the anonymous reviewers for their insightful comments and suggestions, which have greatly improved the manuscript.

## Author contributions

Xingxing Wang: Investigation, data curation, formal analysis, writing - original draft; Mingyang Hu: Data curation, writing - original draft, writing - review & editing; Ying Li: Data curation, writing - original draft; Xiuming Liu: Writing - original draft, resources, funding acquisition; Dean Xiao: Writing - original draft, resources, funding acquisition; Bin Lian: Conceptualization, supervision, project administration, writing - review & editing, resources, funding acquisition.

## Funding

This study was jointly supported by the Science and Technology Support Plan of Guizhou Province (Grant Number 2023–145) and the Major Scientific and Technological Achievement Transformation Project of Guizhou Province (Grant Number 2022-Major-010).

## Declarations

### Competing interests

The authors declare no competing interests.

### Additional information

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-37627-3>.

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